

circuit.

13. The apparatus of claim 10, further comprising:

frequency regulation logic to change an operating frequency of the first integrated circuit, the frequency regulation logic to receive a signal from the program.

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14. The apparatus of claim 10, further comprising:

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voltage regulation logic to change an operating voltage of the first integrated circuit, the voltage regulation logic to receive a signal from the program.

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15. The apparatus of claim 10, wherein the instructions reside in a Basic Input Output System.

16. The apparatus of claim 10, wherein the instructions reside in an operating system.

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17. The apparatus of claim 10, wherein the instructions reside in application software.

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18. The apparatus of claim 10, wherein the first integrated circuit comprises a chip set.

19. The apparatus of claim 10, wherein the first integrated circuit comprises a processing unit.

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20. The apparatus of claim 15, wherein the Basic Input Output System to receive a notification signal from an operating system that the user event has occurred.

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21. The apparatus of claim 14, wherein the program comprises an increasing state transition algorithm discrete from a decreasing state transition algorithm.

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22. The apparatus of claim 10, wherein the program to transition the first integrated circuit to a next higher state of performance based upon an occurrence of a non-user event increases utilization of the first integrated circuit over a preset threshold.

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23. A machine-readable medium that provides instructions, which when executed by a machine, cause the machine to perform operations comprising:

detecting (402) an event to transition an integrated circuit of a computing system to a higher state of performance, the integrated circuit having multiple states of performance including a first state of performance, a second state of performance higher than the first state of performance, and a third state of performance higher than the second state of performance; and

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determining (404) whether the event is a user-initiated event or a software-initiated event; if the event is a user-initiated event directly transitioning (406) the integrated circuit from the first state of performance to the third state of performance based upon detecting the user event, if the event is a software-initiated event transitioning (408) the integrated circuit from the first state of performance to the second state of performance.

24. The machine-readable medium of claim 23, further comprising instructions which, when executed by the machine, cause the machine to perform the further operations comprising:

changing an operating frequency of the integrated circuit to change the state of performance of the integrated circuit.

25. The machine-readable medium of claim 23, further comprising instructions which, when executed by the machine, cause the machine to perform the further operations comprising:

changing an operating voltage level of the integrated circuit to change the state of performance of the integrated circuit.

26. The machine-readable medium of claim 23, further comprising instructions which, when executed by the machine, cause the machine to perform the further operations comprising:

operating the integrated circuit at the third state of performance for a transient period of time.

27. The machine-readable medium of one of claims 23 to 26, further comprising instructions which, when executed by the machine, cause the machine to perform the further operations comprising:

changing the state of performance of the integrated circuit by changing an amount of processors to manage the processing load.

Patentansprüche

1. Verfahren, umfassend:

Detektieren (402) eines Ereignisses, um eine integrierte Schaltung eines Computersystems in einen höheren Performance-Zustand zu versetzen, wobei die integrierte Schaltung mehrere Performance-Zustände aufweist einschließlich eines ersten Performance-Zustandes, eines zweiten Performance-Zustands, der höher als